

Linear Independence

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Observe that, although this way of writing one vector as a combination of the others

$$\vec{s}_0 = c_1 \vec{s}_1 + c_2 \vec{s}_2 + \cdots + c_n \vec{s}_n$$

visually sets off $\vec{s}_0$, algebraically there is nothing special about that vector in that equation. For any $\vec{s}_i$ with a coefficient c_i that is non-0, we can rewrite to isolate $\vec{s}_i$.

$$\vec{s}_i = (1/c_i)\vec{s}_0 + \cdots + (-c_{i-1}/c_i)\vec{s}_{i-1} + (-c_{i+1}/c_i)\vec{s}_{i+1} + \cdots + (-c_n/c_i)\vec{s}_n$$

When we don't want to single out any vector we will instead say that $\vec{s}_0, \vec{s}_1, \dots, \vec{s}_n$ are in a *linear relationship* and put all of the vectors on the same side.

Lemma A subset S of a vector space is linearly independent if and only if among its elements the only linear relationship $c_1 \vec{s}_1 + \cdots + c_n \vec{s}_n = \vec{0}$ is the trivial one, $c_1 = 0, \dots, c_n = 0$ (where $\vec{s}_i \neq \vec{s}_j$ when $i \neq j$).

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Proof If S is linearly independent then no vector $\vec{s}_i$ is a linear combination of other vectors from S , so there is no linear relationship where some of the $\vec{s}$'s have nonzero coefficients.

If S is not linearly independent then some $\vec{s}_i$ is a linear combination $\vec{s}_i = c_1 \vec{s}_1 + \cdots + c_{i-1} \vec{s}_{i-1} + c_{i+1} \vec{s}_{i+1} + \cdots + c_n \vec{s}_n$ of other vectors from S . Subtracting $\vec{s}_i$ from both sides gives a relationship involving a nonzero coefficient, the -1 in front of $\vec{s}_i$. QED

Remark: So to decide if a list of vectors $\vec{s}_0, \dots, \vec{s}_n$ is linearly independent, set up the equation $\vec{0} = c_0 \vec{s}_0 + \cdots + c_n \vec{s}_n$, and calculate whether it has any solutions, other than the trivial one where all coefficients are zero.

Example This set of vectors in the plane $\mathbb{R}^2$ is linearly independent.

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

The only solution to this equation

$$c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

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Example In the vector space of cubic polynomials

$\mathcal{P}_3 = \{a_0 + a_1x + a_2x^2 + a_3x^3 \mid a_i \in \mathbb{R}\}$ the set $\{1 - x, 1 + x\}$ is linearly independent. Setting up the equation $c_0(1 - x) + c_1(1 + x) = 0$ and considering the constant term and linear term, leads to this system

$$\begin{aligned} c_0 + c_1 &= 0 \\ -c_0 + c_1 &= 0 \end{aligned}$$

which has only the trivial solution.

Example The nonzero rows of this matrix form a linearly independent set.

$$\begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 1 & -3 & 1/2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

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Example This subset of $\mathbb{R}^3$ is linearly dependent.

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} \right\}$$

One way to see that is to spot that the third vector is twice the first plus the second. Another way is to solve the linear system

$$c_1 - c_2 + c_3 = 0$$

$$c_1 + c_2 + 3c_3 = 0$$

$$3c_1 \quad + 6c_3 = 0$$

and note that it has more than just the solution $c_1 = c_2 = c_3 = 0$.

Lemma Where V is a vector space, S is a subset of that space, and $\vec{v}$ is an element of that space, $[S \cup \{\vec{v}\}] = [S]$ if and only if $\vec{v} \in [S]$.

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Example The book has the proof; here is an illustration. The span of this set is the xy -plane.

$$P = \left\{ \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \right\} \subseteq \mathbb{R}^3$$

If we expand that by adding a vector to get $\{\vec{p}_1, \vec{p}_2, \vec{q}\}$ then there are two possibilities.

$$P_0 = \left\{ \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \right\} \quad P_1 = \left\{ \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \right\}$$

If the new vector is already in the starting span, $\vec{q} \in [P]$, then the span is unchanged, as with $[P_0] = [P]$. But if the new vector is outside the starting span, $\vec{q} \notin [P]$, then the span grows, as in $[P_1] \supsetneq [P]$.

Corollary For $\vec{v} \in S$, omitting that vector does not shrink the span if and only if that vector is dependent on other vectors in the set. That is, $[S] = [S - \{\vec{v}\}]$ if and only if $\vec{v} \in [S - \{\vec{v}\}]$.

Example These two subsets of $\mathbb{R}^3$ have the same span

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \right\} \quad \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \right\}$$

because in the first set $\vec{v}_3 = 2\vec{v}_2 - \vec{v}_1$.

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because in the first set $\vec{v}_3 = 2\vec{v}_2 - \vec{v}_1$.

Corollary A set S is linearly independent if and only if for any $\vec{v} \in S$, its removal shrinks the span $[S - \{\vec{v}\}] \subsetneq [S]$.

Example This is a linearly independent subset of $\mathcal{P}_3$

$$S = \{1 + x, 1 - x, x^2\}$$

Removal of any element, such as if we remove $1 - x$ to get $\hat{S} = \{1 + x, x^2\}$, will make the span smaller: $[\hat{S}] \subsetneq [S]$.

Lemma Suppose that S is linearly independent and that $\vec{v} \notin S$. Then the set $S \cup \{\vec{v}\}$ is linearly independent if and only if $\vec{v} \notin [S]$.

Example Here is an illustration. Consider this linearly independent subset of $\mathcal{P}_2$.

$$S = \{1 - x, 1 + x\}$$

Its span $[S]$ is the set of linear polynomials $\{a + bx \mid a, b \in \mathbb{R}\}$. (To check: consider $a + bx = r_1(1 - x) + r_2(1 + x)$, which gives a linear system with equations $r_1 + r_2 = a$ and $-r_1 + r_2 = b$, having the solution $r_2 = (1/2)a + (1/2)b$ and $r_1 = (1/2)a - (1/2)b$.)

Here are two supersets.

$$S_1 = S \cup \{2 + 2x\} \quad S_2 = S \cup \{2 + x^2\}$$

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On the left, adding a linear polynomial just adds “repeat information” so $[S_1] = [S]$ and S_1 is linearly dependent.

The right, with “new information,” enlarges the span $[S_2] = \mathcal{P}_2 \supsetneq [S]$ and the new set S_2 is also linearly independent. (To check this, use $a + bx + cx^2 = r_1(1 - x) + r_2(1 + x) + r_3(2 + x^2)$ to get a linear system with solution $r_3 = c$, $r_2 = (1/2)a + (1/2)b$ and $r_1 = (1/2)a - (1/2)b - c$.)

Corollary In a vector space, any finite set has a linearly independent subset with the same span.

Consider the example on the next slide.

Example Consider this subset of $\mathbb{R}^2$.

$$S = \{\vec{s}_1, \vec{s}_2, \vec{s}_3, \vec{s}_4, \vec{s}_5\} = \left\{ \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

The linear relationship

$$r_1 \begin{pmatrix} 2 \\ 2 \end{pmatrix} + r_2 \begin{pmatrix} 3 \\ 3 \end{pmatrix} + r_3 \begin{pmatrix} 1 \\ 4 \end{pmatrix} + r_4 \begin{pmatrix} 0 \\ -1 \end{pmatrix} + r_5 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

gives a system of equations.

$$\begin{array}{rclcl} 2r_1 + 3r_2 + r_3 & + & r_5 & = & 0 \\ 2r_1 + 3r_2 + 4r_3 - r_4 - r_5 & = & 0 \end{array} \xrightarrow{-\rho_1 + \rho_2} \begin{array}{rclcl} 2r_1 + 3r_2 + r_3 & + & r_5 & = & 0 \\ & + & 3r_3 - r_4 - 2r_5 & = & 0 \end{array}$$

Parametrize by expressing the leading variables r_1 and r_3 in terms of the free variables.

$$\left\{ \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{pmatrix} = \begin{pmatrix} -3/2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} r_2 + \begin{pmatrix} -1/6 \\ 0 \\ 1/3 \\ 1 \\ 0 \end{pmatrix} r_4 + \begin{pmatrix} -5/6 \\ 0 \\ 2/3 \\ 0 \\ 1 \end{pmatrix} r_5 \mid r_2, r_4, r_5 \in \mathbb{R} \right\}$$

Parametrize by expressing the leading variables r_1 and r_3 in terms of the free variables.

$$\begin{Bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{Bmatrix} = \begin{Bmatrix} -3/2 \\ 1 \\ 0 \\ 0 \\ 0 \end{Bmatrix} r_2 + \begin{Bmatrix} -1/6 \\ 0 \\ 1/3 \\ 1 \\ 0 \end{Bmatrix} r_4 + \begin{Bmatrix} -5/6 \\ 0 \\ 2/3 \\ 0 \\ 1 \end{Bmatrix} r_5 \mid r_2, r_4, r_5 \in \mathbb{R}$$

Set $r_5 = 1$ and $r_2 = r_4 = 0$ to get $r_1 = -5/6$ and $r_3 = 2/3$,

$$-\frac{5}{6} \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix} + 0 \cdot \begin{pmatrix} 3 \\ 3 \end{pmatrix} + \frac{2}{3} \cdot \begin{pmatrix} 1 \\ 4 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

showing that $\vec{s}_5$ is in the span of the set $\{\vec{s}_1, \vec{s}_3\}$.

Similarly, setting $r_4 = 1$ and the other parameters to 0 shows $\vec{s}_4$ is in the span of the set $\{\vec{s}_1, \vec{s}_3\}$. Also, setting $r_2 = 1$ and the other parameters to 0 shows $\vec{s}_2$ is in the span of the same set.

So without shrinking the span we can omit the vectors $\vec{s}_2$, $\vec{s}_4$, $\vec{s}_5$ associated with the free variables. The set of vectors associated with the leading variables, $\{\vec{s}_1, \vec{s}_3\}$, is linearly independent and so we cannot omit any members without shrinking the span.

Corollary A subset $S = \{\vec{s}_1, \dots, \vec{s}_n\}$ of a vector space is linearly dependent if and only if some $\vec{s}_i$ is a linear combination of the vectors $\vec{s}_1, \dots, \vec{s}_{i-1}$ listed before it.

Corollary A subset $S = \{\vec{s}_1, \dots, \vec{s}_n\}$ of a vector space is linearly dependent if and only if some $\vec{s}_i$ is a linear combination of the vectors $\vec{s}_1, \dots, \vec{s}_{i-1}$ listed before it.

Proof Consider $S_0 = \{\}$, $S_1 = \{\vec{s}_1\}$, $S_2 = \{\vec{s}_1, \vec{s}_2\}$, etc. Some index $i \geq 1$ is the first one with $S_{i-1} \cup \{\vec{s}_i\}$ linearly dependent, and there $\vec{s}_i \in [S_{i-1}]$. QED

Linear independence and subset

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Proof Both are clear.

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This table summarizes the cases.

	$\hat{S} \subset S$	$\hat{S} \supset S$
S independent	$\hat{S}$ must be independent	$\hat{S}$ may be either
S dependent	$\hat{S}$ may be either	$\hat{S}$ must be dependent

An example of the lower left is that the set S of all vectors in the space $\mathbb{R}^2$ is linearly dependent but the subset $\hat{S}$ consisting of only the unit vector on the x -axis is independent. By interchanging $\hat{S}$ with S that's also an example of the upper right.

Independence versus span

The box on the prior slide describes what happens if we start with a linearly independent set S and pass to a new one, $\hat{S}$. If the new set is smaller, $\hat{S} \subset S$, then $\hat{S}$ is sure to be linearly independent. But if the new set is bigger, $S \subset \hat{S}$, then it may be linearly dependent. So, speaking informally, **linearly independent sets like to have fewer elements.**

Contrast that with spanning sets. Suppose that we start with a spanning set S and pass to $\hat{S}$. If the new set is bigger, $S \subset \hat{S}$, then it is sure to be a spanning set. But if the new set is smaller, $\hat{S} \subset S$, then possibly it no longer spans the space. So, again speaking informally, **spanning sets like to have more elements.**

Example This set spans $\mathbb{R}^3$ and is linearly independent.

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

Omit any member and it no longer spans the space. Add any new vector to it and the result is linearly dependent.

Basis and Dimension

Definition of basis

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Example This is a basis for $\mathbb{R}^2$.

$$\left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle$$

It is linearly independent.

$$c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \begin{cases} c_1 + c_2 = 0 \\ -c_1 + c_2 = 0 \end{cases} \implies c_1 = 0, c_2 = 0$$

And it spans $\mathbb{R}^2$ since

$$c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{cases} c_1 + c_2 = x \\ -c_1 + c_2 = y \end{cases}$$

has the solution $c_1 = (1/2)x - (1/2)y$ and $c_2 = (1/2)x + (1/2)y$.

Example In the vector space of linear polynomials $\mathcal{P}_1 = \{a + bx \mid a, b \in \mathbb{R}\}$ one basis is $B = \langle 1 + x, 1 - x \rangle$.

Check that is a basis by verifying that it is linearly independent

$$0 = c_1(1 + x) + c_2(1 - x) \implies 0 = c_1 + c_2, 0 = c_1 - c_2 \implies c_1 = c_2 = 0$$

and that it spans the space.

$$a + bx = c_1(1 + x) + c_2(1 - x) \implies c_1 = (a + b)/2, c_2 = (a - b)/2$$

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$$a + bx = c_1(1 + x) + c_2(1 - x) \implies c_1 = (a + b)/2, \quad c_2 = (a - b)/2$$

Example This is a basis for $\mathcal{M}_{2 \times 2}$.

$$\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix} \right\rangle$$

This is another one.

$$\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right\rangle$$

Example This is a basis for $\mathbb{R}^3$.

$$\mathcal{E}_3 = \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle$$

Calculus books sometimes call those $\vec{i}$, $\vec{j}$, and $\vec{k}$.

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Definition For any $\mathbb{R}^n$

$$\mathcal{E}_n = \left\langle \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\rangle$$

is the *standard* (or *natural*) basis. We denote these vectors $\vec{e}_1, \dots, \vec{e}_n$.

Checking that $\mathcal{E}_n$ is a basis for $\mathbb{R}^n$ is routine.

Theorem In any vector space, a subset is a basis if and only if each vector in the space can be expressed as a linear combination of elements of the subset in one and only one way.

Theorem In any vector space, a subset is a basis if and only if each vector in the space can be expressed as a linear combination of elements of the subset in one and only one way.

Proof A sequence is a basis if and only if its vectors form a set that spans and that is linearly independent. A subset is a spanning set if and only if each vector in the space is a linear combination of elements of that subset in at least one way. Thus we need only show that a spanning subset is linearly independent if and only if every vector in the space is a linear combination of elements from the subset in at most one way.

Consider two expressions of a vector as a linear combination of the members of the subset. Rearrange the two sums, and if necessary add some $0 \cdot \vec{\beta}_i$ terms, so that the two sums combine the same $\vec{\beta}$'s in the same order: $\vec{v} = c_1\vec{\beta}_1 + c_2\vec{\beta}_2 + \cdots + c_n\vec{\beta}_n$ and $\vec{v} = d_1\vec{\beta}_1 + d_2\vec{\beta}_2 + \cdots + d_n\vec{\beta}_n$. Now

$$c_1\vec{\beta}_1 + c_2\vec{\beta}_2 + \cdots + c_n\vec{\beta}_n = d_1\vec{\beta}_1 + d_2\vec{\beta}_2 + \cdots + d_n\vec{\beta}_n$$

holds if and only if

$$(c_1 - d_1)\vec{\beta}_1 + \cdots + (c_n - d_n)\vec{\beta}_n = \vec{0}$$

holds. So, asserting that each coefficient in the lower equation is zero is the same thing as asserting that $c_i = d_i$ for each i , that is, that every vector is expressible as a linear combination of the $\vec{\beta}$'s in a unique way. QED

Example This is a vector and basis for the vector space $\mathbb{R}^3$.

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \in \mathbb{R}^3 \quad B = \left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle = \langle \vec{\beta}_1, \vec{\beta}_2, \vec{\beta}_3 \rangle$$

Find how to express $\vec{v}$ as $c_1\vec{\beta}_1 + c_2\vec{\beta}_2 + c_3\vec{\beta}_3$ by solving this system.

$$\begin{aligned} c_1 + c_2 + c_3 &= 1 \\ c_1 + c_2 &= 2 \\ c_1 &= 3 \end{aligned}$$

By eye we see just what the Theorem says we will see: there is one and only one solution $c_1 = 3$, $c_2 = -1$, and $c_3 = -1$.

Definition In a vector space with basis B the *representation of $\vec{v}$ with respect to B* is the column vector of the coefficients used to express $\vec{v}$ as a linear combination of the basis vectors:

$$\text{Rep}_B(\vec{v}) = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}_B$$

where $B = \langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$ and $\vec{v} = c_1\vec{\beta}_1 + c_2\vec{\beta}_2 + \dots + c_n\vec{\beta}_n$. The c 's are the *coordinates of $\vec{v}$ with respect to B* .

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The prior slide shows that where

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad B = \left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle$$

we have this.

$$\text{Rep}_B(\vec{v}) = \begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix}_B$$

Example Above we saw that in $\mathcal{P}_1 = \{a + bx \mid a, b \in \mathbb{R}\}$ one basis is $B = \langle 1 + x, 1 - x \rangle$. As part of that we computed the coefficients needed to express a member of $\mathcal{P}_1$ as a combination of basis vectors.

$$a + bx = c_1(1 + x) + c_2(1 - x) \implies c_1 = (a + b)/2, c_2 = (a - b)/2$$

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For instance, the polynomial $3 + 4x$ has this expression

$$3 + 4x = (7/2) \cdot (1 + x) + (-1/2) \cdot (1 - x)$$

so its representation is this.

$$\text{Rep}_B(3 + 4x) = \begin{pmatrix} 7/2 \\ -1/2 \end{pmatrix}$$

Example With respect to $\mathbb{R}^3$'s standard basis $\mathcal{E}_3$ the vector

$$\vec{v} = \begin{pmatrix} 2 \\ -3 \\ 1/2 \end{pmatrix}$$

has this representation.

$$\text{Rep}_{\mathcal{E}_3}(\vec{v}) = \begin{pmatrix} 2 \\ -3 \\ 1/2 \end{pmatrix}$$

In general, any $\vec{w} \in \mathbb{R}^n$ has $\text{Rep}_{\mathcal{E}_n}(\vec{w}) = \vec{w}$.

“Represents” in what sense?

The word used in the definition applies because, as this result says, a relationship holds among a set of vectors if and only if it holds among the representatives.

Lemma Where B is a basis with n elements, for any set of vectors, $\alpha_1 \vec{v}_1 + \cdots + \alpha_k \vec{v}_k = \vec{0}_V$ if and only if $\alpha_1 \text{Rep}_B(\vec{v}_1) + \cdots + \alpha_k \text{Rep}_B(\vec{v}_k) = \vec{0}_{\mathbb{R}^n}$.

Example Consider this basis for $\mathbb{R}^2$, and these members of the space.

$$B = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\rangle \quad \vec{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \vec{v}_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Notice that $2\vec{v}_1 - 3\vec{v}_2 - \vec{v}_3 = \vec{0}$. The calculation for these is routine.

$$\text{Rep}_B(\vec{v}_1) = \begin{pmatrix} 5/2 \\ -1/2 \end{pmatrix} \quad \text{Rep}_B(\vec{v}_2) = \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix} \quad \text{Rep}_B(\vec{v}_3) = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

Also routine is the verification that $2 \cdot \text{Rep}_B(\vec{v}_1) - 3 \cdot \text{Rep}_B(\vec{v}_2) - \text{Rep}_B(\vec{v}_3)$ equals $\vec{0}$.

Dimension

Definition of dimension

Definition A vector space is *finite-dimensional* if it has a basis with only finitely many vectors.

Example The space $\mathbb{R}^3$ is finite-dimensional since it has a basis with three elements $\mathcal{E}_3$.

Example The space of quadratic polynomials $\mathcal{P}_2$ has at least one basis with finitely many elements, $\langle 1, x, x^2 \rangle$, so it is finite-dimensional.

Example The space $\mathcal{M}_{2 \times 2}$ of 2×2 matrices is finite-dimensional. Here is one basis with finitely many members.

$$\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right\rangle$$

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Note From this point on we will restrict our attention to vector spaces that are finite-dimensional. All the later examples, definitions, and theorems assume this of the spaces.

We will show that for any finite-dimensional space, all of its bases have the same number of elements.

Example Each of these is a basis for $\mathcal{P}_2$.

$$B_0 = \langle 1, 1 + x, 1 + x + x^2 \rangle$$

$$B_1 = \langle 1 + x + x^2, 1 + x, 1 \rangle$$

$$B_2 = \langle x^2, 1 + x, 1 - x \rangle$$

$$B_3 = \langle 1, x, x^2 \rangle$$

Each has three elements.

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Each has three elements.

Example Here are two different bases for $\mathcal{M}_{2 \times 2}$.

$$B_0 = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right\rangle$$

$$B_1 = \left\langle \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right\rangle$$

Both have four elements.

All of a space's bases are the same size

Theorem In any finite-dimensional vector space, all bases have the same number of elements.

Proof Fix a vector space with at least one finite basis. From among all of this space's bases, choose one $B = \langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$ that has minimal size. We will show that any other basis $D = \langle \vec{\delta}_1, \vec{\delta}_2, \dots \rangle$ also has n members. Because B has minimal size, D cannot have fewer than n vectors. We will argue that it cannot have more.

So suppose that $\vec{\delta}_1, \dots, \vec{\delta}_{n+1}$ are distinct. We shall show that among these $n + 1$ vectors there is a nontrivial linear relationship, a $\alpha_1 \vec{\delta}_1 + \dots + \alpha_{n+1} \vec{\delta}_{n+1} = \vec{0}$ where not all the α_i are zero.

First represent them with respect to B .

$$\text{Rep}_B(\vec{\delta}_1) = \begin{pmatrix} c_{1,1} \\ \vdots \\ c_{n,1} \end{pmatrix} \quad \cdots \quad \text{Rep}_B(\vec{\delta}_{n+1}) = \begin{pmatrix} c_{1,n+1} \\ \vdots \\ c_{n,n+1} \end{pmatrix}$$

Previous lemma says that a relationship holds among the vectors $\alpha_1 \vec{\delta}_1 + \cdots + \alpha_{n+1} \vec{\delta}_{n+1} = \vec{0}$ if and only if the same relationship holds among the representations.

$$\alpha_1 \begin{pmatrix} c_{1,1} \\ \vdots \\ c_{n,1} \end{pmatrix} + \cdots + \alpha_{n+1} \begin{pmatrix} c_{1,n+1} \\ \vdots \\ c_{n,n+1} \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

The $c_{i,j}$ are the coefficients in the representations of the given vectors, while we are looking for the α_i . So this gives a homogeneous linear system.

$$\begin{array}{c} c_{1,1} \alpha_1 + \cdots + c_{1,n+1} \alpha_{n+1} = 0 \\ \vdots \\ c_{n,1} \alpha_1 + \cdots + c_{n,n+1} \alpha_{n+1} = 0 \end{array}$$

It is homogeneous so it has at least one solution. But there are more unknowns, $n + 1$, than equations. So it has infinitely many solutions. Therefore the set D is linearly dependent.

QED

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Example The vector space $\mathcal{M}_{n \times m}$ has dimension $n \cdot m$. A natural basis consists of matrices with a single 1 and the other entries 0's.

Example The solution set S of this system

$$\begin{aligned}x - y + z &= 0 \\ -x + 2y - z + 2w &= 0 \\ -x + 3y - z + 4w &= 0\end{aligned}$$

is a vector space (this is easy to check for any homogeneous system).
Solving the system

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & 0 & 0 \\ -1 & 2 & -1 & 2 & 0 \\ 1 & 3 & -1 & 4 & 0 \end{array} \right) \xrightarrow[\rho_1 + \rho_3]{\rho_1 + \rho_2} \xrightarrow{-2\rho_2 + \rho_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

and parametrizing gives a basis of two vectors.

$$\left\{ \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \cdot z + \begin{pmatrix} -2 \\ -2 \\ 0 \\ 1 \end{pmatrix} \cdot w \mid z, w \in \mathbb{R} \right\} \quad B = \left\langle \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \\ 0 \\ 1 \end{pmatrix} \right\rangle$$

So S is a vector space of dimension two.

Corollary No linearly independent set can have a size greater than the dimension of the enclosing space.

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Corollary In an n -dimensional space, a set composed of n vectors is linearly independent if and only if it spans the space.

Example This clearly spans the space.

$$\left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle \subseteq \mathbb{R}^3$$

Because it has same number of elements as the dimension of the space, it is therefore a basis.

Vector Spaces and Linear Systems

Row space

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Lemma If two matrices A and B are related by a row operation

$$A \xrightarrow{\rho_i \leftrightarrow \rho_j} B \quad \text{or} \quad A \xrightarrow{k\rho_i} B \quad \text{or} \quad A \xrightarrow{k\rho_i + \rho_j} B$$

(for $i \neq j$ and $k \neq 0$) then their row spaces are equal. Hence, row-equivalent matrices have the same row space and therefore the same row rank.

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(for $i \neq j$ and $k \neq 0$) then their row spaces are equal. Hence, row-equivalent matrices have the same row space and therefore the same row rank.

Proof When $A \rightarrow B$ then each row of B is a linear combination of the rows of A. That is, in the above terminology, each row of B is an element of the row space of A. Then $\text{Rowspace}(B) \subseteq \text{Rowspace}(A)$ follows because a member of the set $\text{Rowspace}(B)$ is a linear combination of the rows of B, so it is a combination of combinations of the rows of A, and by the Linear Combination Lemma is also a member of $\text{Rowspace}(A)$.

For the other set containment, recall that **row operations are reversible** so $A \rightarrow B$ if and only if $B \rightarrow A$. Then $\text{Rowspace}(A) \subseteq \text{Rowspace}(B)$ follows as in the previous paragraph. QED

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Lemma The nonzero rows of an echelon form matrix make up a linearly independent set.

Proof Recall that no nonzero row of an echelon form matrix is a linear combination of the other rows. QED

Example The matrix before Gauss's Method and the matrix after have equal row spaces.

$$M = \begin{pmatrix} 1 & 2 & 1 & 0 & 3 \\ -1 & -2 & 2 & 2 & 0 \\ 2 & 4 & 5 & 2 & 9 \end{pmatrix} \xrightarrow[\begin{smallmatrix} \rho_1 + \rho_2 \\ -2\rho_1 + \rho_3 \end{smallmatrix}]{-\rho_2 + \rho_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 3 \\ 0 & 0 & 3 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The nonzero rows of the latter matrix form a basis for $\text{Rowspace}(M)$.

$$B = \langle (1 \ 2 \ 1 \ 0 \ 3), (0 \ 0 \ 3 \ 2 \ 3) \rangle$$

The row rank is 2.

So Gauss's Method produces a basis for the row space of a matrix. It has found the "**repeat/redundant**" information, that M 's third row is three times the first plus the second, and eliminated that extra row.

Column space

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Example This system

$$\begin{aligned}2x + 3y &= d_1 \\ -x + (1/2)y &= d_2\end{aligned}$$

has a solution for those $d_1, d_2 \in \mathbb{R}$ that we can find to satisfy this vector equation.

$$x \cdot \begin{pmatrix} 2 \\ -1 \end{pmatrix} + y \cdot \begin{pmatrix} 3 \\ 1/2 \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \quad x, y \in \mathbb{R}$$

That is, the system has a solution if and only if the vector on the right is in the column space of this matrix.

$$\begin{pmatrix} 2 & 3 \\ -1 & 1/2 \end{pmatrix}$$

Transpose

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Example To find a basis for the column space of a matrix,

$$\begin{pmatrix} 2 & 3 \\ -1 & 1/2 \end{pmatrix}$$

transpose,

$$\begin{pmatrix} 2 & 3 \\ -1 & 1/2 \end{pmatrix}^T = \begin{pmatrix} 2 & -1 \\ 3 & 1/2 \end{pmatrix}$$

reduce,

$$\begin{pmatrix} 2 & -1 \\ 3 & 1/2 \end{pmatrix} \xrightarrow{(-3/2)\rho_1 + \rho_2} \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$$

and transpose back.

$$\begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}^T = \begin{pmatrix} 2 & 0 \\ -1 & 2 \end{pmatrix}$$

This basis

$$B = \left\langle \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\rangle$$

shows that the column space is the entire vector space $\mathbb{R}^2$.

Lemma Row operations do not change the column rank.

The reason is that row operations do not change linear relationships between the columns. So row operations do not change the number of linearly unrelated columns.

Proof as exercise. Here is an example. In this matrix, the second column minus the first is equal to the third.

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 3 \\ 2 & 6 & 4 \end{pmatrix}$$

As we perform row operations that relationship between the columns continues to hold.

$$\xrightarrow{-2\rho_1 + \rho_3} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \end{pmatrix} \xrightarrow{-(2/3)\rho_2 + \rho_3} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

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Proof Bring the matrix to reduced echelon form. Then the row rank equals the number of leading entries since that equals the number of nonzero rows. Then also, the number of leading entries equals the column rank because the set of columns containing leading entries consists of some of the $\vec{e}_i$'s from a standard basis, and that set is linearly independent and spans the set of columns. Hence, in the reduced echelon form matrix, the row rank equals the column rank, because each equals the number of leading entries.

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Definition The *rank* of a matrix is its row rank or column rank.

Example The column rank of this matrix

$$\begin{pmatrix} 2 & -1 & 3 & 1 & 0 & 1 \\ 3 & 0 & 1 & 1 & 4 & -1 \\ 4 & -2 & 6 & 2 & 0 & 2 \\ 1 & 0 & 3 & 0 & 0 & 2 \end{pmatrix}$$

is 3. Its largest set of linearly independent columns is size 3 because that's the size of its largest set of linearly independent rows.

$$\begin{array}{l} -(3/2)\rho_1 + \rho_2 \\ -2\rho_1 + \rho_3 \\ -(1/2)\rho_1 + \rho_4 \end{array} \quad \begin{array}{l} -(1/3)\rho_2 + \rho_4 \\ \rho_3 \leftrightarrow \rho_4 \end{array} \quad \begin{pmatrix} 2 & -1 & 3 & 1 & 0 & 1 \\ 0 & 3/2 & -7/2 & -1/2 & 4 & -5/2 \\ 0 & 0 & 8/3 & -1/3 & -4/3 & 7/3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

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- (1) the rank of A is r
- (2) the vector space of solutions of the associated homogeneous system has dimension $n - r$

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Proof The rank of A is r if and only if Gaussian reduction on A ends with r nonzero rows. That's true if and only if echelon form matrices row equivalent to A have r -many leading variables. That in turn holds if and only if there are $n - r$ free variables. QED

Corollary Where the matrix A is $n \times n$, these statements

- (1) the rank of A is n
- (2) A is nonsingular
- (3) the rows of A form a linearly independent set
- (4) the columns of A form a linearly independent set
- (5) any linear system whose matrix of coefficients is A has one and only one solution

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are equivalent.

Proof Clearly (1) $\iff$ (2) $\iff$ (3) $\iff$ (4). The last, (4) $\iff$ (5), holds because a set of n column vectors is linearly independent if and only if it is a basis for $\mathbb{R}^n$, but the system

$$c_1 \begin{pmatrix} a_{1,1} \\ a_{2,1} \\ \vdots \\ a_{m,1} \end{pmatrix} + \cdots + c_n \begin{pmatrix} a_{1,n} \\ a_{2,n} \\ \vdots \\ a_{m,n} \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_m \end{pmatrix}$$

has a unique solution for all choices of $d_1, \dots, d_m \in \mathbb{R}$ if and only if the vectors of a 's on the left form a basis.

QED

An AI Example: Basis of Face Images



Figure: AT&T Faces Dataset

An AI Example: Basis of Face Images

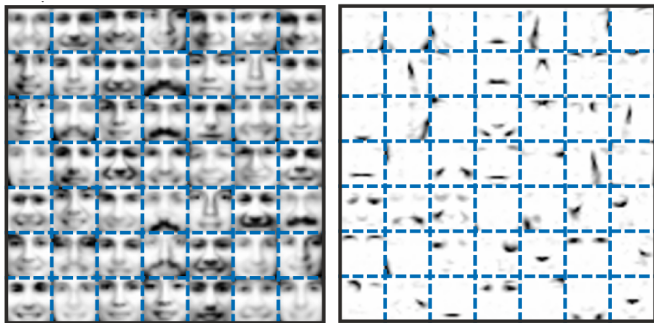


Figure: Two sets of basis for face images. Any image $\vec{v}$ can be expressed by the basis.
 $\vec{v} = \sum_{i=1}^n c_i \vec{\beta}_i$. Image Source: Lee and Seung Nature (1999)

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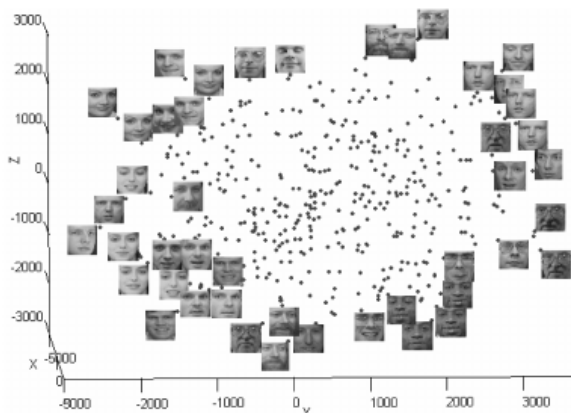


Figure: Coordinates of Images in 3D space. Similar images have similar coordinates.

Isomorphism

Example We have the intuition that the vector spaces $\mathbb{R}^2$ and $\mathcal{P}_1$ are “the same,” in that they are two-component spaces. For instance

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ is just like } 1 + 2x,$$

$$\text{and } \begin{pmatrix} -3 \\ 1/2 \end{pmatrix} \text{ is just like } -3 + (1/2)x,$$

etc. What makes the spaces alike, not just the sets, is that the association persists through the operations: this illustrates addition

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ 1/2 \end{pmatrix} = \begin{pmatrix} -2 \\ 5/2 \end{pmatrix}$$

$$\text{is just like } (1 + 2x) + (-3 + (1/2)x) = -2 + (5/2)x$$

and this illustrates scalar multiplication.

$$3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \text{ is just like } 3(1 + 2x) = 3 + 6x$$

Example Similarly, we can link each two-tall vector with a linear polynomial.

$$\begin{pmatrix} a \\ b \end{pmatrix} \longleftrightarrow a + bx$$

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This association holds through the vector space operations of addition

$$\begin{pmatrix} a_1 \\ b_1 \end{pmatrix} + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \end{pmatrix} \\ \longleftrightarrow (a_1 + b_1x) + (a_2 + b_2x) = (a_1 + a_2) + (b_1 + b_2)x$$

and scalar multiplication.

$$r \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ra \\ rb \end{pmatrix} \longleftrightarrow r(a + bx) = (ra) + (rb)x$$

We say that the association *preserves the structure* of the spaces.

Example We can think of $\mathcal{M}_{2 \times 2}$ as “the same” as $\mathbb{R}^4$ if we associate in this way.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longleftrightarrow \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

For instance, these are corresponding elements.

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This association persists under addition.

$$\begin{aligned} \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} + \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} &= \begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{pmatrix} \\ &\longleftrightarrow \begin{pmatrix} a_1 \\ b_1 \\ c_1 \\ d_1 \end{pmatrix} + \begin{pmatrix} a_2 \\ b_2 \\ c_2 \\ d_2 \end{pmatrix} = \begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \\ c_1 + c_2 \\ d_1 + d_2 \end{pmatrix} \end{aligned}$$

Here is an example of addition being preserved under this association.

$$\begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} + \begin{pmatrix} 0 & 4 \\ 3 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 5 & -5 \end{pmatrix}$$

$$\longleftrightarrow \begin{pmatrix} 1 \\ -1 \\ 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \\ 3 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 5 \\ -5 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} + \begin{pmatrix} 0 & 4 \\ 3 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 5 & -5 \end{pmatrix} \quad \longleftrightarrow \quad \begin{pmatrix} 1 \\ -1 \\ 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \\ 3 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 5 \\ -5 \end{pmatrix}$$

The association also persists through scalar multiplication.

$$r \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ra & rb \\ rc & rd \end{pmatrix} \quad \longleftrightarrow \quad r \cdot \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} ra \\ rb \\ rc \\ rd \end{pmatrix}$$

This illustrates.

$$2 \cdot \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 4 & -4 \end{pmatrix} \quad \longleftrightarrow \quad 2 \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 4 \\ -4 \end{pmatrix}$$

Isomorphism

Definition

An *isomorphism* between two vector spaces V and W is a map $f: V \rightarrow W$ that

- 1) is a correspondence: f is one-to-one and onto;
- 2) *preserves structure*: if $\vec{v}_1, \vec{v}_2 \in V$ then

$$f(\vec{v}_1 + \vec{v}_2) = f(\vec{v}_1) + f(\vec{v}_2)$$

and if $\vec{v} \in V$ and $r \in \mathbb{R}$ then

$$f(r\vec{v}) = rf(\vec{v})$$

(we write $V \cong W$, read “ V is isomorphic to W ”, when such a map exists).

How-to

To verify that $f: V \rightarrow W$ is an isomorphism, do these four.

- ▶ To show that f is one-to-one, assume that $\vec{v}_1, \vec{v}_2 \in V$ are such that $f(\vec{v}_1) = f(\vec{v}_2)$ and derive that $\vec{v}_1 = \vec{v}_2$.
- ▶ To show that f is onto, let $\vec{w}$ be an element of W and find a $\vec{v} \in V$ such that $f(\vec{v}) = \vec{w}$.
- ▶ To show that f preserves addition, check that for all $\vec{v}_1, \vec{v}_2 \in V$ we have $f(\vec{v}_1 + \vec{v}_2) = f(\vec{v}_1) + f(\vec{v}_2)$.
- ▶ To show that f preserves scalar multiplication, check that for all $\vec{v} \in V$ and $r \in \mathbb{R}$ we have $f(r \cdot \vec{v}) = r \cdot f(\vec{v})$.

The first two cover condition (1), that the spaces correspond, that for each member of W there exactly one associated member of V . The latter two cover (2), that the map preserves structure. For these two, the intuition is in the discussion above. (Later section cover these two at length.)

Example The space of quadratic polynomials $\mathcal{P}_2$ is isomorphic to $\mathbb{R}^3$ under this map.

$$f(a_0 + a_1x + a_2x^2) = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$$

Here are two examples of the action of f .

$$f(1 + 2x + 3x^2) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad f(3 + 4x^2) = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$$

To verify that f is an isomorphism we must check condition (1), that f is a correspondence, and condition (2), that f preserves structure.

The first part of the definition's clause (1) is that f is one-to-one. We suppose $f(\vec{v}_1) = f(\vec{v}_2)$, that the function yields the same output on two inputs $f(a_0 + a_1x + a_2x^2) = f(b_0 + b_1x + b_2x^2)$. From that, we must derive that the two inputs are equal. The definition of f gives

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix}$$

and two column vectors are equal only if their entries are equal $a_0 = b_0$, $a_1 = b_1$, and $a_2 = b_2$. Thus the original inputs are equal $\vec{v}_1 = a_0 + a_1x + a_2x^2 = b_0 + b_1x + b_2x^2 = \vec{v}_2$. So f is one-to-one.

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The second part of (1) is that f is onto. We consider an element of the codomain

$$\vec{w} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \in \mathbb{R}^3$$

and produce an element of the domain that maps to it. Observe that $\vec{w}$ is the image under f of the member $\vec{v} = a_0 + a_1x + a_2x^2$ of the domain. Thus f is onto.

The definition's clause (2) also has two halves. First we show that f preserves addition. Consider f acting on the sum of two elements of the domain.

$$\begin{aligned} f((a_0 + a_1x + a_2x^2) + (b_0 + b_1x + b_2x^2)) \\ = f((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2) \end{aligned}$$

The definition of f gives

$$= \begin{pmatrix} a_0 + b_0 \\ a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}$$

and that equals

$$= \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix}$$

which gives

$$= f(a_0 + a_1x + a_2x^2) + f(b_0 + b_1x + b_2x^2)$$

as required.

We finish by checking that f preserves scalar multiplication. This is similar to the check for addition.

$$\begin{aligned} f(r \cdot (a_0 + a_1x + a_2x^2)) &= f((ra_0) + (ra_1)x + (ra_2)x^2) \\ &= \begin{pmatrix} ra_0 \\ ra_1 \\ ra_2 \end{pmatrix} \\ &= r \cdot \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \\ &= r \cdot f(a_0 + a_1x + a_2x^2) \end{aligned}$$

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So the function f is an isomorphism. Because there is an isomorphism, the two spaces are isomorphic $\mathcal{P}_2 \cong \mathbb{R}^3$.

Example Consider these two vector spaces (under the natural operations)

$$V = \left\{ \begin{pmatrix} a & b \\ c & 0 \end{pmatrix} \mid a, b, c \in \mathbb{R} \right\} \quad W = \{ (x \ y \ z) \mid x, y, z \in \mathbb{R} \}$$

and consider this function.

$$\begin{pmatrix} a & b \\ c & 0 \end{pmatrix} \xrightarrow{f} (b \ 2a \ a + c)$$

Here is an example of the map's action.

$$\begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix} \xrightarrow{f} (2 \ 6 \ 4)$$

We will verify that f is an isomorphism.

To show that f is one-to-one, suppose that

$$f\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & 0 \end{pmatrix}\right) = f\left(\begin{pmatrix} a_2 & b_2 \\ c_2 & 0 \end{pmatrix}\right)$$

Then $(b_1 - 2a_1 \quad a_1 + c_1) = (b_2 - 2a_2 \quad a_2 + c_2)$. The first entries give that $b_1 = b_2$, the second entries that $a_1 = a_2$, and with that the third entries give that $c_1 = c_2$.

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & 0 \end{pmatrix} = \begin{pmatrix} a_2 & b_2 \\ c_2 & 0 \end{pmatrix}$$

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Then $(b_1 - 2a_1 - a_1 + c_1) = (b_2 - 2a_2 - a_2 + c_2)$. The first entries give that $b_1 = b_2$, the second entries that $a_1 = a_2$, and with that the third entries give that $c_1 = c_2$.

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & 0 \end{pmatrix} = \begin{pmatrix} a_2 & b_2 \\ c_2 & 0 \end{pmatrix}$$

To show that f is onto, consider a member of W .

$$\vec{w} = (x \ y \ z)$$

We must find a $\vec{v}$ so that $f(\vec{v}) = \vec{w}$. The map sends the upper right entry of the input to the first entry of the output, so the upper right of $\vec{v}$ is x . Similarly, the upper left of $\vec{v}$ is $(1/2)y$. With that, the lower left is $z - (1/2)y$.

$$(x \ y \ z) = f\left(\begin{pmatrix} y/2 & x \\ z - y/2 & 0 \end{pmatrix}\right)$$

To show that f preserves addition, assume

$$f\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & 0 \end{pmatrix} + \begin{pmatrix} a_2 & b_2 \\ c_2 & 0 \end{pmatrix}\right) = f\left(\begin{pmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & 0 \end{pmatrix}\right)$$

which equals $(b_1 + b_2 \quad 2(a_1 + a_2) \quad (a_1 + a_2) + (c_1 + c_2))$. In turn, that equals this.

$$(b_1 \quad 2a_1 \quad a_1 + c_1) + (b_2 \quad 2a_2 \quad a_2 + c_2) = f\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & 0 \end{pmatrix}\right) + f\left(\begin{pmatrix} a_2 & b_2 \\ c_2 & 0 \end{pmatrix}\right)$$

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which equals $(b_1 + b_2 \ 2(a_1 + a_2) \ (a_1 + a_2) + (c_1 + c_2))$. In turn, that equals this.

$$(b_1 \ 2a_1 \ a_1 + c_1) + (b_2 \ 2a_2 \ a_2 + c_2) = f\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & 0 \end{pmatrix}\right) + f\left(\begin{pmatrix} a_2 & b_2 \\ c_2 & 0 \end{pmatrix}\right)$$

Preservation of scalar multiplication is similar.

$$\begin{aligned} f\left(r \cdot \begin{pmatrix} a & b \\ c & 0 \end{pmatrix}\right) &= f\left(\begin{pmatrix} ra & rb \\ rc & 0 \end{pmatrix}\right) \\ &= (rb \ 2ra \ ra + rc) \\ &= r \cdot (b \ 2a \ a + c) \\ &= r \cdot f\left(\begin{pmatrix} a & b \\ c & 0 \end{pmatrix}\right) \end{aligned}$$

Preservation is special

Many functions do not preserve addition and scalar multiplication. For instance, $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$$

does not preserve addition since the sum done one way

$$f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix}\right) = f\left(\begin{pmatrix} 3 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 9 \\ 0 \end{pmatrix}$$

gives a different result than the sum done the other way.

$$f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) + f\left(\begin{pmatrix} 2 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$

Special case: Automorphisms

Definition

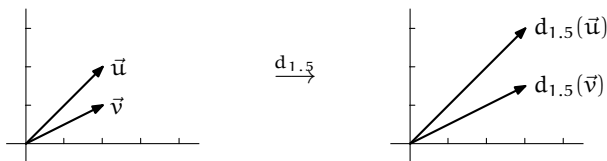
An *automorphism* is an isomorphism of a space with itself.

Special case: Automorphisms

Definition

An *automorphism* is an isomorphism of a space with itself.

Example A *dilation* map $d_s: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that multiplies all vectors by a nonzero scalar s is an automorphism of $\mathbb{R}^2$.

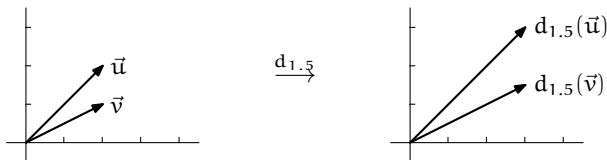


Special case: Automorphisms

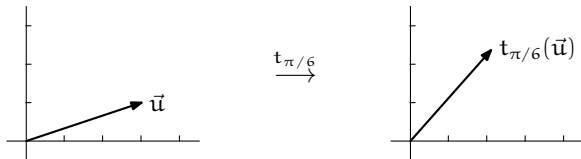
Definition

An **automorphism** is an isomorphism of a space with itself.

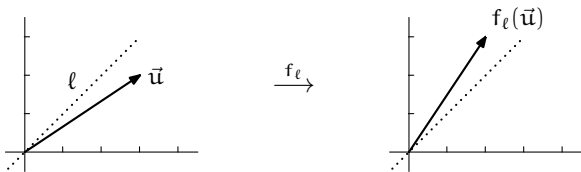
Example A **dilation** map $d_s: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that multiplies all vectors by a nonzero scalar s is an automorphism of $\mathbb{R}^2$.



Another automorphism is a **rotation** or **turning map**, $t_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates all vectors through an angle θ .

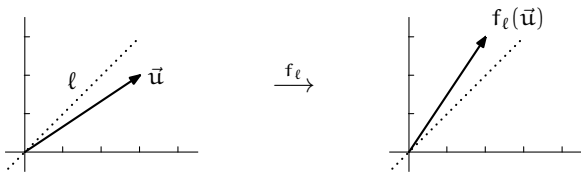


A third type of automorphism of $\mathbb{R}^2$ is a map $f_\ell: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that *flips* or *reflects* all vectors over a line ℓ through the origin.



Checking that each is an isomorphism is an exercise.

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Checking that each is an isomorphism is an exercise.

Why study automorphisms? Isn't it trivial that the plane is just like itself?

Consider the family of automorphisms t_θ rotating all vectors counterclockwise. They make precise the intuition that the plane is uniform—that space near the x-axis is just like space near the y-axis.

So one lesson is that we can use maps to describe relationships between spaces. If the maps are isomorphisms then this relation makes precise the intuition “just like”.

A second lesson is that while there is an obvious automorphism of $\mathbb{R}^2$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix}$$

there are reasons to consider maps other than the obvious one.

Lemma An isomorphism maps a zero vector to a zero vector.

Lemma An isomorphism maps a zero vector to a zero vector.

Proof Where $f: V \rightarrow W$ is an isomorphism, fix some $\vec{v} \in V$. Then
 $f(\vec{0}_V) = f(0 \cdot \vec{v}) = 0 \cdot f(\vec{v}) = \vec{0}_W$.

QED

Lemma For any map $f: V \rightarrow W$ between vector spaces these statements are equivalent.

(1) f preserves structure

$$f(\vec{v}_1 + \vec{v}_2) = f(\vec{v}_1) + f(\vec{v}_2) \quad \text{and} \quad f(c\vec{v}) = c f(\vec{v})$$

(2) f preserves linear combinations of two vectors

$$f(c_1\vec{v}_1 + c_2\vec{v}_2) = c_1 f(\vec{v}_1) + c_2 f(\vec{v}_2)$$

(3) f preserves linear combinations of any finite number of vectors

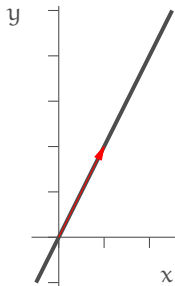
$$f(c_1\vec{v}_1 + \cdots + c_n\vec{v}_n) = c_1 f(\vec{v}_1) + \cdots + c_n f(\vec{v}_n)$$

The book contains the details of both proofs.

Remark Examination of the proofs show that they depend only on clause (2) of the definition, not on that the map is a correspondence. We will say more in the next section.

Example The line through the origin with slope two is a vector space.

$$L = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \mid y = 2x \right\} = \left\{ \begin{pmatrix} t \\ 2t \end{pmatrix} \mid t \in \mathbb{R} \right\}$$



The parametrization

$$L = \left\{ \begin{pmatrix} t \\ 2t \end{pmatrix} \mid t \in \mathbb{R} \right\} = \left\{ t \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

suggests that L is just like $\mathbb{R}$: there is a point on L associated with $1 \in \mathbb{R}$, a point associated with $2 \in \mathbb{R}$, etc. We will show that this function is an isomorphism between L and $\mathbb{R}^1$.

$$f\left(\begin{pmatrix} t \\ 2t \end{pmatrix}\right) = t$$

To verify that f is one-to-one suppose that f maps two members of L to the same output.

$$f\left(t_1 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = f\left(t_2 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right)$$

By the definition of f we have $t_1 = t_2$ and so the two input members of L are equal.

To check that f is onto consider a member of the codomain, $r \in \mathbb{R}$. There is a member of the domain L that maps to it, namely this one.

$$f\left(r \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = r$$

To finish, we combine the two structure checks, using the lemma's (2).

$$\begin{aligned} f\left(c_1 \cdot \begin{pmatrix} t_1 \\ 2t_1 \end{pmatrix} + c_2 \cdot \begin{pmatrix} t_2 \\ 2t_2 \end{pmatrix}\right) &= f\left((c_1 t_1 + c_2 t_2) \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) \\ &= c_1 t_1 + c_2 t_2 = c_1 \cdot f\left(\begin{pmatrix} t_1 \\ 2t_1 \end{pmatrix}\right) + c_2 \cdot f\left(\begin{pmatrix} t_2 \\ 2t_2 \end{pmatrix}\right) \end{aligned}$$

Dimension characterizes isomorphism

Lemma The inverse of an isomorphism is also an isomorphism.

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Proof Suppose that V is isomorphic to W via $f: V \rightarrow W$. An isomorphism is a correspondence between the sets so f has an inverse function $f^{-1}: W \rightarrow V$ that is also a correspondence.

Lemma The inverse of an isomorphism is also an isomorphism.

Proof Suppose that V is isomorphic to W via $f: V \rightarrow W$. An isomorphism is a correspondence between the sets so f has an inverse function $f^{-1}: W \rightarrow V$ that is also a correspondence.

We will show that because f preserves linear combinations, so also does f^{-1} . Suppose that $\vec{w}_1, \vec{w}_2 \in W$. Because it is an isomorphism, f is onto and there are $\vec{v}_1, \vec{v}_2 \in V$ such that $\vec{w}_1 = f(\vec{v}_1)$ and $\vec{w}_2 = f(\vec{v}_2)$. Then

$$\begin{aligned} f^{-1}(c_1 \cdot \vec{w}_1 + c_2 \cdot \vec{w}_2) &= f^{-1}(c_1 \cdot f(\vec{v}_1) + c_2 \cdot f(\vec{v}_2)) \\ &= f^{-1}(f(c_1 \vec{v}_1 + c_2 \vec{v}_2)) = c_1 \vec{v}_1 + c_2 \vec{v}_2 = c_1 \cdot f^{-1}(\vec{w}_1) + c_2 \cdot f^{-1}(\vec{w}_2) \end{aligned}$$

since $f^{-1}(\vec{w}_1) = \vec{v}_1$ and $f^{-1}(\vec{w}_2) = \vec{v}_2$. With that, by Lemma 1.11's second statement, this map preserves structure. QED

Example We saw earlier that this line through the origin subspace of $\mathbb{R}^2$

$$L = \left\{ \begin{pmatrix} t \\ 2t \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

is isomorphic to $\mathbb{R}^1$ via this function.

$$f\left(t \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = t$$

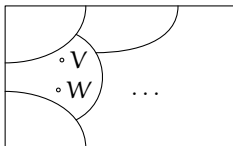
The inverse $f^{-1}: \mathbb{R} \rightarrow L$ given by

$$f^{-1}(r) = r \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} r \\ 2r \end{pmatrix}$$

is also an isomorphism.

Theorem Isomorphism is an equivalence relation between vector spaces.

The book contains the proof; here is a diagram of what it tells us: the collection of all finite-dimensional vector spaces is partitioned into classes. Two spaces are in the same class if they are isomorphic.



The next result characterizes these classes.

Theorem Vector spaces are isomorphic if and only if they have the same dimension.

The proof is these two lemmas.

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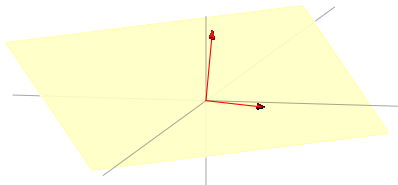
The proof is these two lemmas.

Lemma If spaces are isomorphic then they have the same dimension.

Lemma If spaces have the same dimension then they are isomorphic.

The proof of the first depends on showing that under a isomorphism the image of a domain basis is a range basis. The proof of the second depends on giving an explicit isomorphism from an n -dimensional domain to $\mathbb{R}^n$. See the book for the full proofs; here we will illustrate with an example.

Example The plane $2x - y + z = 0$ through the origin in $\mathbb{R}^3$ is a vector space (under the natural operations).



Describe the space as a span by taking that to be a one-equation linear system and parametrizing $x = (1/2)y - (1/2)z$.

$$P = \left\{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} z \mid y, z \in \mathbb{R} \right\}$$

Clearly the set with those two vectors is linearly independent so it is a basis.

$$B = \left\langle \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \right\rangle$$

Here is the basis from the prior slide.

$$B = \left\langle \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \right\rangle$$

Because its basis has two vectors, P is a dimension 2 space.

The second lemma's proof shows that this is an isomorphism: the map $f: P \rightarrow \mathbb{R}^2$ that associates each element $\vec{v} \in P$ with its representation $\text{Rep}_B(\vec{v}) \in \mathbb{R}^2$.

To illustrate the association, here is an example of its action on an $\mathbb{R}^2$ vector picked at random.

$$\vec{v}_1 = \begin{pmatrix} 7/2 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} \cdot 3 + \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \cdot (-4) \quad \xrightarrow{f} \quad \text{Rep}_B(\vec{v}_1) = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

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Another (as with the prior one, $\vec{v}_2$ is picked at random from $\mathbb{R}^2$).

$$\vec{v}_2 = \begin{pmatrix} -17/4 \\ 1/2 \\ 9 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} \cdot (1/2) + \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \cdot 9 \xrightarrow{f} \text{Rep}_B(\vec{v}_2) = \begin{pmatrix} 1/2 \\ 9 \end{pmatrix}$$

The first lemma's proof shows that any isomorphism takes bases to bases: starting with basis vectors $\vec{\beta}_i$ for the domain and applying an isomorphism f gives basis vectors $f(\vec{\beta}_i)$ for the range.

For the isomorphism from the prior slide we have

$$\vec{\beta}_1 = \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} \cdot 1 + \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \cdot 0 \xrightarrow{f} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

and

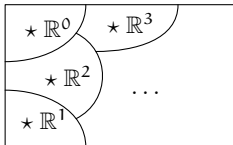
$$\vec{\beta}_2 = \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} \cdot 0 + \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \cdot 1 \xrightarrow{f} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

which together make the basis $\mathcal{E}_2$ for $\mathbb{R}^2$.

Corollary Each finite-dimensional vector space is isomorphic to one and only one of the $\mathbb{R}^n$.

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Thus the real spaces $\mathbb{R}^n$ form a set of canonical representatives of the isomorphism classes—every isomorphism class contains one and only one $\mathbb{R}^n$.



Homomorphism

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Definition A function between vector spaces $h: V \rightarrow W$ that preserves addition

$$\text{if } \vec{v}_1, \vec{v}_2 \in V \text{ then } h(\vec{v}_1 + \vec{v}_2) = h(\vec{v}_1) + h(\vec{v}_2)$$

and scalar multiplication

$$\text{if } \vec{v} \in V \text{ and } r \in \mathbb{R} \text{ then } h(r \cdot \vec{v}) = r \cdot h(\vec{v})$$

is a *homomorphism* or *linear map*.

Example Of these two maps $h, g: \mathbb{R}^2 \rightarrow \mathbb{R}$, the first is a homomorphism while the second is not.

$$\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{h} 2x - 3y \qquad \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{g} 2x - 3y + 1$$

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The map h respects addition

$$\begin{aligned} h\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) &= h\left(\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}\right) = 2(x_1 + x_2) - 3(y_1 + y_2) \\ &= (2x_1 - 3y_1) + (2x_2 - 3y_2) = h\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}\right) + h\left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) \end{aligned}$$

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$$r \cdot h\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = r \cdot (2x - 3y) = 2rx - 3ry = (2r)x - (3r)y = h\left(r \cdot \begin{pmatrix} x \\ y \end{pmatrix}\right)$$

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In contrast, g does not respect addition.

$$g\left(\begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 5 \\ 6 \end{pmatrix}\right) = -17 \qquad g\left(\begin{pmatrix} 1 \\ 4 \end{pmatrix}\right) + g\left(\begin{pmatrix} 5 \\ 6 \end{pmatrix}\right) = -16$$

We proved these two while studying isomorphisms.

Lemma A linear map sends the zero vector to the zero vector.

Lemma The following are equivalent for any map $f: V \rightarrow W$ between vector spaces.

- (1) f is a homomorphism
- (2) $f(c_1 \cdot \vec{v}_1 + c_2 \cdot \vec{v}_2) = c_1 \cdot f(\vec{v}_1) + c_2 \cdot f(\vec{v}_2)$ for any $c_1, c_2 \in \mathbb{R}$ and $\vec{v}_1, \vec{v}_2 \in V$
- (3) $f(c_1 \cdot \vec{v}_1 + \cdots + c_n \cdot \vec{v}_n) = c_1 \cdot f(\vec{v}_1) + \cdots + c_n \cdot f(\vec{v}_n)$ for any $c_1, \dots, c_n \in \mathbb{R}$ and $\vec{v}_1, \dots, \vec{v}_n \in V$

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Example Between any two vector spaces the zero map

$Z: V \rightarrow W$ given by $Z(\vec{v}) = \vec{0}_W$ is a linear map. Using (2):

$$Z(c_1 \vec{v}_1 + c_2 \vec{v}_2) = \vec{0}_W = \vec{0}_W + \vec{0}_W = c_1 Z(\vec{v}_1) + c_2 Z(\vec{v}_2).$$

Example The *inclusion map* $\iota: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

$$\iota\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

is a homomorphism.

$$\begin{aligned} \iota\left(c_1 \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + c_2 \cdot \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) &= \iota\left(\begin{pmatrix} c_1 x_1 + c_2 x_2 \\ c_1 y_1 + c_2 y_2 \end{pmatrix}\right) \\ &= \begin{pmatrix} c_1 x_1 + c_2 x_2 \\ c_1 y_1 + c_2 y_2 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} c_1 x_1 \\ c_1 y_1 \\ 0 \end{pmatrix} + \begin{pmatrix} c_2 x_2 \\ c_2 y_2 \\ 0 \end{pmatrix} \\ &= c_1 \cdot \iota\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}\right) + c_2 \cdot \iota\left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) \end{aligned}$$

Example The derivative is a transformation on polynomial spaces. For instance, consider $d/dx: \mathcal{P}_2 \rightarrow \mathcal{P}_1$ given by

$$d/dx (ax^2 + bx + c) = 2ax + b$$

(examples are $d/dx (3x^2 - 2x + 4) = 6x - 2$ and $d/dx (x^2 + 1) = 2x$).

It is a homomorphism.

$$\begin{aligned} d/dx (r_1(a_1x^2 + b_1x + c_1) + r_2(a_2x^2 + b_2x + c_2)) \\ &= d/dx ((r_1a_1 + r_2a_2)x^2 + (r_1b_1 + r_2b_2)x + (r_1c_1 + r_2c_2)) \\ &= 2(r_1a_1 + r_2a_2)x + (r_1b_1 + r_2b_2) \\ &= (2r_1a_1x + r_1b_1) + (2r_2a_2x + r_2b_2) \\ &= r_1 \cdot d/dx (a_1x^2 + b_1x + c_1) + r_2 \cdot d/dx (a_2x^2 + b_2x + c_2) \end{aligned}$$

Example The *trace* of a square matrix is the sum down the upper-left to lower-right diagonal. Thus $\text{Tr}: \mathcal{M}_{2 \times 2} \rightarrow \mathbb{R}$ is this.

$$\text{Tr}\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = a + d$$

It is linear.

$$\begin{aligned} \text{Tr}\left(r_1 \cdot \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} + r_2 \cdot \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right) \\ = \text{Tr}\left(\begin{pmatrix} r_1 a_1 + r_2 a_2 & r_1 b_1 + r_2 b_2 \\ r_1 c_1 + r_2 c_2 & r_1 d_1 + r_2 d_2 \end{pmatrix}\right) \\ = (r_1 a_1 + r_2 a_2) + (r_1 d_1 + r_2 d_2) \\ = r_1(a_1 + d_1) + r_2(a_2 + d_2) \\ = r_1 \cdot \text{Tr}\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\right) + r_2 \cdot \text{Tr}\left(\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right) \end{aligned}$$

Theorem A homomorphism is determined by its action on a basis: if V is a vector space with basis $\langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$, if W is a vector space, and if $\vec{w}_1, \dots, \vec{w}_n \in W$ (these codomain elements need not be distinct) then there exists a homomorphism from V to W sending each $\vec{\beta}_i$ to $\vec{w}_i$, and that homomorphism is unique.

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Example Proof as exercise. Here is an illustration. Consider a map $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with this action on a basis.

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{h} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{h} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

The effect of the map on any vector $\vec{v}$ at all is determined by those two facts. Represent that vector $\vec{v}$ with respect to the basis.

$$\begin{pmatrix} -1 \\ 5 \end{pmatrix} = 5 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 6 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Compute $h(\vec{v})$ using the definition of homomorphism.

$$h(\vec{v}) = h\left(5 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 6 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = 5 \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} - 6 \cdot \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -7 \\ 15 \end{pmatrix}$$

Example Consider $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with this effect on the standard basis.

$$f(\vec{e}_1) = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \quad f(\vec{e}_2) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad f(\vec{e}_3) = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}$$

Because this is the standard basis, the effect of the map on any vector $\vec{v} \in \mathbb{R}^3$ is especially easy to compute. For instance,

$$\text{Rep}_{\mathcal{E}_3, \mathcal{E}_3} \left(\begin{pmatrix} -5 \\ 0 \\ 10 \end{pmatrix} \right) = \begin{pmatrix} -5 \\ 0 \\ 10 \end{pmatrix}$$

and so we have this.

$$f\left(\begin{pmatrix} -5 \\ 0 \\ 10 \end{pmatrix}\right) = -5 \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 10 \cdot \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 25 \\ -20 \\ 5 \end{pmatrix}$$

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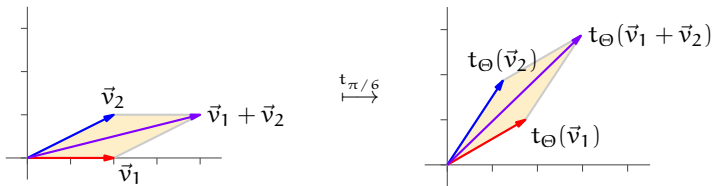
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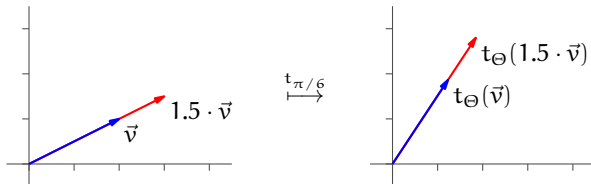
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Definition Let V and W be vector spaces and let $B = \langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$ be a basis for V . A function defined on that basis $f: B \rightarrow W$ is *extended linearly* to a function $\hat{f}: V \rightarrow W$ if for all $\vec{v} \in V$ such that $\vec{v} = c_1 \vec{\beta}_1 + \dots + c_n \vec{\beta}_n$, the action of the map is $\hat{f}(\vec{v}) = c_1 \cdot f(\vec{\beta}_1) + \dots + c_n \cdot f(\vec{\beta}_n)$.

Example Consider the action $t_\Theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ of rotating all vectors in the plane through an angle Θ . These drawings show that this map satisfies the addition



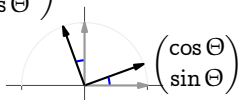
and scalar multiplication conditions.



We will develop the formula for t_Θ .

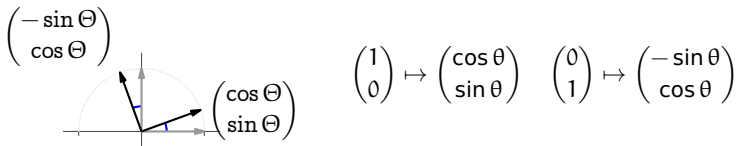
Fix a basis for the domain $\mathbb{R}^2$; the standard basis $\mathcal{E}_2$ is convenient. We want the basis vectors mapped as here.

$$\begin{pmatrix} -\sin \Theta \\ \cos \Theta \end{pmatrix}$$



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

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Extend linearly.

$$\begin{aligned} t_\theta \left(\begin{pmatrix} x \\ y \end{pmatrix} \right) &= t_\theta \left(x \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= x \cdot t_\theta \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) + y \cdot t_\theta \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= x \cdot \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} + y \cdot \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix} \end{aligned}$$

Example One basis of the space of quadratic polynomials $\mathcal{P}_2$ is $B = \langle x^2, x, 1 \rangle$. Define the *evaluation map* $\text{eval}_3: \mathcal{P}_2 \rightarrow \mathbb{R}$ by specifying its action on that basis

$$x^2 \xrightarrow{\text{eval}_3} 9 \quad x \xrightarrow{\text{eval}_3} 3 \quad 1 \xrightarrow{\text{eval}_3} 1$$

and then extending linearly.

$$\begin{aligned} \text{eval}_3(ax^2 + bx + c) &= a \cdot \text{eval}_3(x^2) + b \cdot \text{eval}_3(x) + c \cdot \text{eval}_3(1) \\ &= 9a + 3b + c \end{aligned}$$

For instance, $\text{eval}_3(x^2 + 2x + 3) = 9 + 6 + 3 = 18$.

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For instance, $\text{eval}_3(x^2 + 2x + 3) = 9 + 6 + 3 = 18$.

On the basis elements, we can describe the action of this map as: plugging the value 3 in for x . That remains true when we extend linearly, so $\text{eval}_3(p(x)) = p(3)$.

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Example In $\mathbb{R}^3$ the function f_{yz} that reflects vectors over the yz -plane

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \xrightarrow{f_{yz}} \begin{pmatrix} -x \\ y \\ z \end{pmatrix}$$

is a linear transformation.

$$\begin{aligned} f_{yz}\left(r_1 \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + r_2 \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}\right) &= f_{yz}\left(\begin{pmatrix} r_1 x_1 + r_2 x_2 \\ r_1 y_1 + r_2 y_2 \\ r_1 z_1 + r_2 z_2 \end{pmatrix}\right) = \begin{pmatrix} -(r_1 x_1 + r_2 x_2) \\ r_1 y_1 + r_2 y_2 \\ r_1 z_1 + r_2 z_2 \end{pmatrix} \\ &= r_1 \begin{pmatrix} -x_1 \\ y_1 \\ z_1 \end{pmatrix} + r_2 \begin{pmatrix} -x_2 \\ y_2 \\ z_2 \end{pmatrix} = r_1 f_{yz}\left(\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}\right) + r_2 f_{yz}\left(\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}\right) \end{aligned}$$

?? *Lemma* For vector spaces V and W , the set of linear functions from V to W is itself a vector space, a subspace of the space of all functions from V to W .

We denote the space of linear maps from V to W by $\mathcal{L}(V, W)$.

The book contains the proof; instead we will do a couple of examples.

Example We can combine the two homomorphisms $f, g: \mathcal{P}_1 \rightarrow \mathbb{R}^2$

$$f(a_0 + a_1 x) = \begin{pmatrix} a_0 + a_1 \\ 0 \end{pmatrix} \quad g(a_0 + a_1 x) = \begin{pmatrix} 4a_1 \\ a_1 \end{pmatrix}$$

into a function $2f + 3g: \mathcal{P}_1 \rightarrow \mathbb{R}^2$ whose action is this.

$$(2f + 3g)(a_0 + a_1 x) = \begin{pmatrix} 2a_0 + 14a_1 \\ 3a_1 \end{pmatrix}$$

The point of the lemma is that the linear combination $2f + 3g$ is also a homomorphism; the check is routine. The collection of homomorphisms from $\mathcal{P}_1$ to $\mathbb{R}^2$ is closed under linear combinations of those homomorphisms—it is a vector space.

Example Consider $\mathcal{L}(\mathbb{R}, \mathbb{R}^2)$. A member of $\mathcal{L}(\mathbb{R}, \mathbb{R}^2)$ is a linear map. A linear map is determined by its action on a basis of the domain space. Fix these bases.

$$B_{\mathbb{R}} = \mathcal{E}_1 = \langle 1 \rangle \quad B_{\mathbb{R}^2} = \mathcal{E}_2 = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$$

Thus the functions that are elements of $\mathcal{L}(\mathbb{R}, \mathbb{R}^2)$ are determined by c_1 and c_2 here.

$$1 \xrightarrow{t} c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

We could write each such map as $h = h_{c_1, c_2}$. There are two parameters and thus $\mathcal{L}(\mathbb{R}, \mathbb{R}^2)$ is a dimension 2 space.

Range space and null space

Lemma Under a homomorphism, the image of any subspace of the domain is a subspace of the codomain. In particular, the image of the entire space, the range of the homomorphism, is a subspace of the codomain.

The book has the proof; we instead consider an example.

Example Let $f: \mathbb{R}^2 \rightarrow \mathcal{M}_{2 \times 2}$ be

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} a & a+b \\ 2b & b \end{pmatrix}$$

(the check that it is a homomorphism is routine). One subspace of the domain is the x axis.

$$S = \left\{ \begin{pmatrix} a \\ 0 \end{pmatrix} \mid a \in \mathbb{R} \right\}$$

The image under f of the x axis is a subspace of of the codomain $\mathcal{M}_{2 \times 2}$.

$$f(S) = \left\{ \begin{pmatrix} a & a \\ 0 & 0 \end{pmatrix} \mid a \in \mathbb{R} \right\}$$

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Another subspace of $\mathbb{R}^2$ is $\mathbb{R}^2$ itself. The image of $\mathbb{R}^2$ under f is this subspace of $\mathcal{M}_{2 \times 2}$.

$$f(\mathbb{R}^2) = \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \cdot c_1 + \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \cdot c_2 \mid c_1, c_2 \in \mathbb{R} \right\}$$

Example For any angle θ , the function $t_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates vectors counterclockwise through an angle θ is a homomorphism.

In the domain $\mathbb{R}^2$ each line through the origin is a subspace. The image of that line under this map is another line through the origin, a subspace of the codomain $\mathbb{R}^2$.

Range space

Definition The *range space* of a homomorphism $h: V \rightarrow W$ is

$$\mathcal{R}(h) = \{h(\vec{v}) \mid \vec{v} \in V\}$$

sometimes denoted $h(V)$. The dimension of the range space is the map's *rank*.

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Example This map from $\mathcal{M}_{2 \times 2}$ to $\mathbb{R}^2$ is linear.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a + b \\ 2a + 2b \end{pmatrix}$$

The range space is a line through the origin.

$$\left\{ \begin{pmatrix} t \\ 2t \end{pmatrix} \mid t \in \mathbb{R} \right\}$$

Every member of that set is the image of a 2×2 matrix.

$$\begin{pmatrix} t \\ 2t \end{pmatrix} = h\left(\begin{pmatrix} t & 0 \\ 0 & 0 \end{pmatrix}\right)$$

The map's rank is 1.

Example The derivative map $d/dx: \mathcal{P}_4 \rightarrow \mathcal{P}_4$ is linear. Its range is $\mathcal{R}(d/dx) = \mathcal{P}_3$. (Verifying that every member of $\mathcal{P}_3$ is the derivative of some member of $\mathcal{P}_4$ is easy.) The rank of this derivative function is the dimension of $\mathcal{P}_3$, namely 4.

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Example Projection $\pi: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix}$$

is a linear map; the check is routine. The range space is $\mathcal{R}(\pi) = \mathbb{R}^2$ because given a vector $\vec{w} \in \mathbb{R}^2$

$$\vec{w} = \begin{pmatrix} a \\ b \end{pmatrix}$$

we can find a $\vec{v} \in \mathbb{R}^3$ that maps to it, specifically any $\vec{v}$ with a first component a and second component b . Thus the rank of π is 2.

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Example Projection $\pi: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

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is a linear map; the check is routine. The range space is $\mathcal{R}(\pi) = \mathbb{R}^2$ because given a vector $\vec{w} \in \mathbb{R}^2$

$$\vec{w} = \begin{pmatrix} a \\ b \end{pmatrix}$$

we can find a $\vec{v} \in \mathbb{R}^3$ that maps to it, specifically any $\vec{v}$ with a first component a and second component b . Thus the rank of π is 2.

In the book's next section, on computing linear maps, we will do more examples of determining the range space.

Many-to-one

In moving from isomorphisms to homomorphisms we dropped the requirement that the maps be onto and one-to-one. But any homomorphism $h: V \rightarrow W$ is onto its range space $\mathcal{R}(h)$, so dropping the onto condition has, in a way, no effect on the range. It doesn't allow any essentially new maps.

Many-to-one

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In contrast, consider the effect of dropping the one-to-one condition. With that, an output vector $\vec{w} \in W$ may have many associated inputs, many $\vec{v} \in V$ such that $h(\vec{v}) = \vec{w}$.

Recall that for any function $h: V \rightarrow W$, the set of elements of V that map to $\vec{w} \in W$ is the *inverse image* $h^{-1}(\vec{w}) = \{\vec{v} \in V \mid h(\vec{v}) = \vec{w}\}$.

The structure of the inverse image sets will give us insight into the definition of homomorphism.

Example Projection $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}$ onto the x axis is linear.

$$\pi\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = x$$

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Here are some elements of $\pi^{-1}(2)$. Think of these as “2 vectors.”



Think of elements of $\pi^{-1}(3)$ as “3 vectors.”



These elements of $\pi^{-1}(5)$ are “5 vectors.”



These drawings give us a way to make the definition of homomorphism more concrete. Consider preservation of addition.

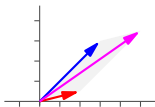
$$\pi(\vec{u}) + \pi(\vec{v}) = \pi(\vec{u} + \vec{v})$$

If $\vec{u}$ is such that $\pi(\vec{u}) = 2$, and $\vec{v}$ is such that $\pi(\vec{v}) = 3$, then $\vec{u} + \vec{v}$ will be such that the sum $\pi(\vec{u} + \vec{v}) = 5$.

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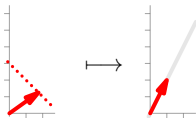


A similar interpretation holds for preservation of scalar multiplication: the image of an “ $r \cdot 2$ vector” is r times 2.

Example This function $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear.

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + y \\ 2x + 2y \end{pmatrix}$$

Here are elements of $h^{-1}\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right)$. (Only one inverse image element is shown as a vector, most are indicated with dots.)



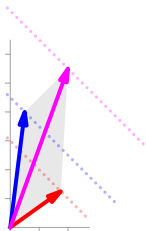
Here are some elements of $h^{-1}\left(\begin{pmatrix} 1.5 \\ 3 \end{pmatrix}\right)$ and $h^{-1}\left(\begin{pmatrix} 2.5 \\ 5 \end{pmatrix}\right)$.



The way that the range space vectors add

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1.5 \\ 3 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 5 \end{pmatrix}$$

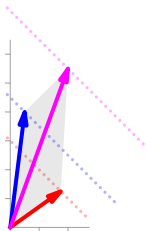
is reflected in the domain: red plus blue makes magenta.



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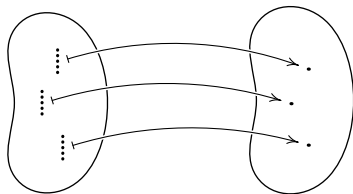
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That is, preservation of addition is: $h(\vec{v}_1) + h(\vec{v}_2) = h(\vec{v}_1 + \vec{v}_2)$.

Homomorphisms organize the domain

So the intuition is that a linear map organizes its domain into inverse images,

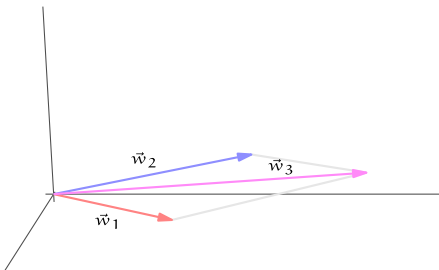


such that those sets reflect the structure of the range.

Example Projection $\pi: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a homomorphism.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix}$$

Here we draw the range $\mathbb{R}^2$ as the xy -plane inside of $\mathbb{R}^3$.

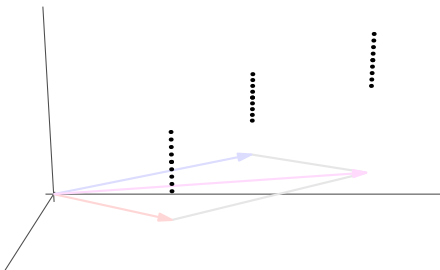


In the range the parallelogram shows a vector addition $\vec{w}_1 + \vec{w}_2 = \vec{w}_3$.

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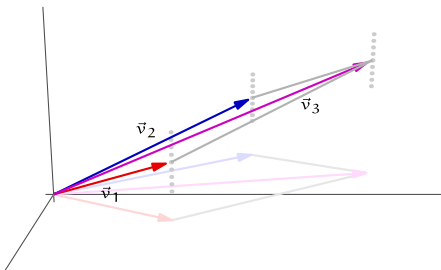
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The diagram shows some of the points in each inverse image $\pi^{-1}(\vec{w}_1)$, $\pi^{-1}(\vec{w}_2)$, and $\pi^{-1}(\vec{w}_3)$.

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The diagram shows some of the points in each inverse image $\pi^{-1}(\vec{w}_1)$, $\pi^{-1}(\vec{w}_2)$, and $\pi^{-1}(\vec{w}_3)$. The sum of a vector $\vec{v}_1 \in \pi^{-1}(\vec{w}_1)$ and a vector $\vec{v}_2 \in \pi^{-1}(\vec{w}_2)$ equals a vector $\vec{v}_3 \in \pi^{-1}(\vec{w}_3)$. A $\vec{w}_1$ vector plus a $\vec{w}_2$ vector equals a $\vec{w}_3$ vector.

This interpretation of the definition of homomorphism also holds when the spaces are not ones that we can sketch.

Example Let $h: \mathcal{P}_2 \rightarrow \mathbb{R}^2$ be

$$ax^2 + bx + c \mapsto \begin{pmatrix} b \\ c \end{pmatrix}$$

and consider these three members of the range such that $\vec{w}_1 + \vec{w}_2 = \vec{w}_3$

$$\vec{w}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \vec{w}_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad \vec{w}_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

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The inverse image of $\vec{w}_1$ is $h^{-1}(\vec{w}_1) = \{a_1x^2 + 1x + c_1 \mid a_1, c_1 \in \mathbb{R}\}$.

Members of this set are “ $\vec{w}_1$ vectors.”

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This interpretation of the definition of homomorphism also holds when the spaces are not ones that we can sketch.

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Any $\vec{v}_1 \in h^{-1}(\vec{w}_1)$ plus any $\vec{v}_2 \in h^{-1}(\vec{w}_2)$ equals a $\vec{v}_3 \in h^{-1}(\vec{w}_3)$: a quadratic with an x coefficient of 1 plus a quadratic with an x coefficient of -1 equals a quadratic with an x coefficient of 0.

Null space

In each of those examples, the homomorphism $h: V \rightarrow W$ shows how to view the domain V as organized into the inverse images $h^{-1}(\vec{w})$.

In the examples these inverse images are all the same, but shifted. So if we describe one of them then we understand how the domain is divided. Vector spaces have a distinguished element, $\vec{0}$. So we next consider the inverse image $h^{-1}(\vec{0})$.

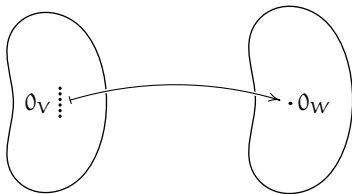
Lemma For any homomorphism the inverse image of a subspace of the range is a subspace of the domain. In particular, the inverse image of the trivial subspace of the range is a subspace of the domain.

The book has the verification.

Definition The *null space* or *kernel* of a linear map $h: V \rightarrow W$ is the inverse image of $\vec{0}_W$.

$$\mathcal{N}(h) = h^{-1}(\vec{0}_W) = \{\vec{v} \in V \mid h(\vec{v}) = \vec{0}_W\}$$

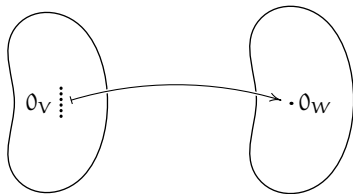
The dimension of the null space is the map's *nullity*.



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The dimension of the null space is the map's **nullity**.



Note Strictly, the trivial subspace of the codomain is not $\vec{0}_W$, it is $\{\vec{0}_W\}$, and so we may think to write the nullspace as $h^{-1}(\{\vec{0}_W\})$. But we have defined the two sets $h^{-1}(\vec{w})$ and $h^{-1}(\{\vec{w}\})$ to be equal and the first is easier to write.

Example Consider the derivative $d/dx: \mathcal{P}_2 \rightarrow \mathcal{P}_1$. This is the nullspace; note that it is a subset of the domain

$$\mathcal{N}(d/dx) = \{ax^2 + bx + c \mid 2ax + b = 0\}$$

(the '0' there is the zero polynomial $0x + 0$). Now, $2ax + b = 0$ if and only if they have the same constant coefficient $b = 0$, the same x coefficient of $a = 0$, and the same coefficient of x^2 (which gives no restriction). So this is the nullspace, and the nullity is 1.

$$\mathcal{N}(d/dx) = \{ax^2 + bx + c \mid a = 0, b = 0, c \in \mathbb{R}\} = \{c \mid c \in \mathbb{R}\}$$

Example The function $h: \mathbb{R}^2 \rightarrow \mathbb{R}^1$ given by

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto 2a + b$$

has this null space and so its nullity is 1.

$$\mathcal{N}(h) = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \mid 2a + b = 0 \right\} = \left\{ \begin{pmatrix} -1/2 \\ 1 \end{pmatrix} b \mid b \in \mathbb{R} \right\}$$

Example The homomorphism $f: \mathcal{M}_{2 \times 2} \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a + b \\ c + d \end{pmatrix}$$

has this null space

$$\begin{aligned} \mathcal{N}(f) &= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a + b = 0 \text{ and } c + d = 0 \right\} \\ &= \left\{ \begin{pmatrix} -b & b \\ -d & d \end{pmatrix} \mid b, d \in \mathbb{R} \right\} \end{aligned}$$

and a nullity of 2.

Example The dilation function $d_3: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} 3a \\ 3b \end{pmatrix}$$

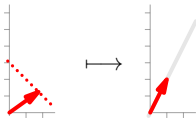
has $\mathcal{N}(d_3) = \{\vec{0}\}$. A trivial space has an empty basis so d_3 's nullity is 0.

Rank plus nullity

Recall the example map $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + y \\ 2x + 2y \end{pmatrix}$$

whose range space $\mathcal{R}(h)$ is the line $y = 2x$ and whose domain is organized into lines, $\mathcal{N}(h)$ is the line $y = -x$. There, an entire line's worth of domain vectors collapses to the single range point.



In moving from domain to range, this map drops a dimension. We can account for it by thinking that each output point absorbs a one-dimensional set.

Theorem A linear map's rank plus its nullity equals the dimension of its domain.

The book contains the proof.

Example Consider this map $h: \mathbb{R}^3 \rightarrow \mathbb{R}$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto x/2 + y/5 + z$$

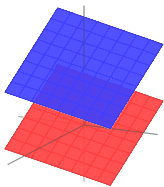
The null space is this plane.

$$\mathcal{N}(h) = h^{-1}(0) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x/2 + y/5 + z = 0 \right\}$$

Other inverse image sets are also planes.

$$h^{-1}(1) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x/2 + y/5 + z = 1 \right\} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid z = 1 - x/2 - y/5 \right\}$$

This shows the inverse images $h^{-1}(0)$ and $h^{-1}(1)$ lined up on the z axis.



So h breaks the domain into stacked planes—any two inverse images $h^{-1}(r_1)$ and $h^{-1}(r_2)$ are collections of domain vectors whose endpoints form a plane. The only difference between these 2-dimensional subsets is where they sit in the stack, shown here as where they intersect z axis.

That is, h partitions the 3-dimensional domain into 2-dimensional sets, leaving 1 dimension of freedom, which matches the dimension of the map's range.

Example Projection $\pi: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \mapsto \begin{pmatrix} a \\ b \end{pmatrix}$$

takes a 3-dimensional domain to a 2-dimensional range. Its null space is the z -axis, so its nullity is 1.

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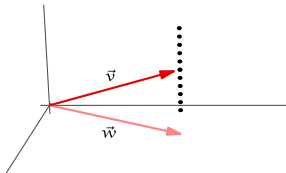
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This example shows the idea of the proof particularly clearly. Take the basis $B_N = \langle \vec{e}_3 \rangle$ for the null space. Expand that to the basis $\mathcal{E}_3$ for the entire domain. On an input vector the action of π is

$$c_1 \vec{e}_1 + c_2 \vec{e}_2 + c_3 \vec{e}_3 \mapsto c_1 \vec{e}_1 + c_2 \vec{e}_2 + \vec{0}$$

and so the domain is organized by π into inverse images that are vertical lines, one-dimensional sets like the null space.



Example The derivative function $d/dx: \mathcal{P}_2 \rightarrow \mathcal{P}_1$

$$ax^2 + bx + c \mapsto 2a \cdot x + b$$

has this range space

$$\mathcal{R}(d/dx) = \{d \cdot x + e \mid d, e \in \mathbb{R}\} = \mathcal{P}_1$$

(the linear polynomial $dx + e \in \mathcal{P}_1$ is the image of any antiderivative $(d/2)x^2 + ex + C$, where $C \in \mathbb{R}$). This is its null space.

$$\mathcal{N}(d/dx) = \{0x^2 + 0x + c \mid c \in \mathbb{R}\} = \{c \mid c \in \mathbb{R}\}$$

The rank is 2 while the nullity is 1, and they add to the domain's dimension 3.

Example The dilation function $d_3: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} 3a \\ 3b \end{pmatrix}$$

has range space $\mathbb{R}^2$ and a trivial nullspace $\mathcal{N}(d_3) = \{\vec{0}\}$. So its rank is 2 and its nullity is 0.

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The book's next section is on computing linear maps, and we will compute more null spaces there.

Lemma Under a linear map, the image of a linearly dependent set is linearly dependent.

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Proof Suppose that $c_1\vec{v}_1 + \cdots + c_n\vec{v}_n = \vec{0}_V$ with some c_i nonzero. Apply h to both sides: $h(c_1\vec{v}_1 + \cdots + c_n\vec{v}_n) = c_1h(\vec{v}_1) + \cdots + c_nh(\vec{v}_n)$ and $h(\vec{0}_V) = \vec{0}_W$. Thus we have $c_1h(\vec{v}_1) + \cdots + c_nh(\vec{v}_n) = \vec{0}_W$ with some c_i nonzero. QED

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Example The trace function $\text{Tr}: \mathcal{M}_{2 \times 2} \rightarrow \mathbb{R}$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a + d$$

is linear. This set of matrices is dependent.

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \right\}$$

The three matrices map to 1, 0, and 2 respectively. The set $\{1, 0, 2\} \subseteq \mathbb{R}$ is linearly dependent.

A one-to-one homomorphism is an isomorphism

Theorem Where V is an n -dimensional vector space, these are equivalent statements about a linear map $h: V \rightarrow W$.

- (1) h is one-to-one
- (2) h has an inverse from its range to its domain that is a linear map
- (3) $\mathcal{N}(h) = \{\vec{0}\}$, that is, $\text{nullity}(h) = 0$
- (4) $\text{rank}(h) = n$
- (5) if $\langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$ is a basis for V then $\langle h(\vec{\beta}_1), \dots, h(\vec{\beta}_n) \rangle$ is a basis for $\mathcal{R}(h)$

The book has the proof.

Transformations of $\mathbb{R}^2$

Lines go to lines

In a real space $\mathbb{R}^n$ a line through the origin is a set $\{r \cdot \vec{v} \mid r \in \mathbb{R}\}$ of multiples of a nonzero vector.

Consider a transformation $t: \mathbb{R}^n \rightarrow \mathbb{R}^n$. It is linear and so t 's action

$$r \cdot \vec{v} \xrightarrow{t} r \cdot t(\vec{v})$$

sends members of the line $\{r \cdot \vec{v} \mid r \in \mathbb{R}\}$ in the domain to members of the line $\{s \cdot t(\vec{v}) \mid s \in \mathbb{R}\}$ in the codomain.

Thus, under a transformation, lines through the origin map to lines through the origin. Further, the action of t is determined by its effect $t(\vec{v})$ on any nonzero element of the domain line.

Example Consider the line $y = 2x$ in the plane

$$\left\{ r \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} \mid r \in \mathbb{R} \right\}$$

and this transformation.

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + 3y \\ 2x + 4y \end{pmatrix}$$

The map's effect on any vector in the line is easy to compute.

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \xrightarrow{t} \begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

The linear map property $t(r \cdot \vec{v}) = r \cdot t(\vec{v})$ imposes a uniformity on t 's action: t has twice the effect on $2\vec{v}$, three times the effect on $3\vec{v}$, etc.

$$\begin{pmatrix} 2 \\ 4 \end{pmatrix} \xrightarrow{t} \begin{pmatrix} 14 \\ 20 \end{pmatrix} \quad \begin{pmatrix} -3 \\ -6 \end{pmatrix} \xrightarrow{t} \begin{pmatrix} -21 \\ -30 \end{pmatrix} \quad \begin{pmatrix} r \\ 2r \end{pmatrix} \xrightarrow{t} \begin{pmatrix} 7r \\ 10r \end{pmatrix}$$

In short: the action of t on any nonzero $\vec{v}$ determines its action on any other vector $r\vec{v}$ in the line $[\{\vec{v}\}]$.

Pick one, any one

Every plane vector is in some line through the origin so to understand what $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ does to plane elements it suffices to understand what it does to lines through the origin.

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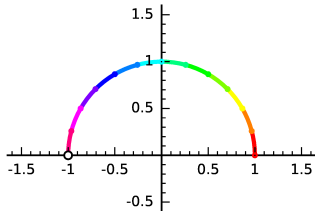
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So one way to understand a transformation's action is to take a set containing one nonzero vector from each line through the origin, and describe where the transformation maps the elements of that set.

A natural set with one nonzero element from each line through the origin is the upper half unit circle (we will explain the colors below).

$$\left\{ \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix} \mid 0 \leq t < \pi \right\}$$

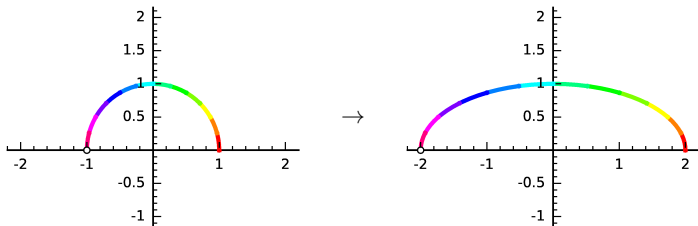


Dilate x

Example The map

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x \\ y \end{pmatrix}$$

doubles the first coordinate while keeping the second coordinate constant. This shows the transformation of the upper half circle.

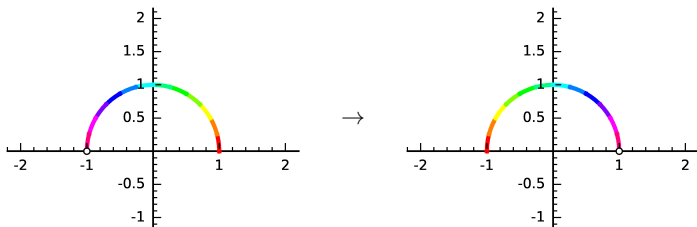


Reverse orientation

Example Here we dilate by a negative.

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} -x \\ y \end{pmatrix}$$

The transformation of the upper half circle shows why we used the colors. In the domain they are, taken counterclockwise, red, orange, yellow, green, blue, indigo, violet. In the codomain, again taken counterclockwise, they do the opposite.

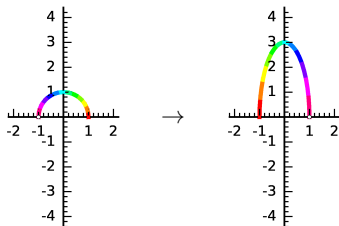


Combine dilations

Example Here we dilate both x and y .

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} -x \\ 3y \end{pmatrix}$$

Again the color order reverses, in addition to the stretching along the y axis.

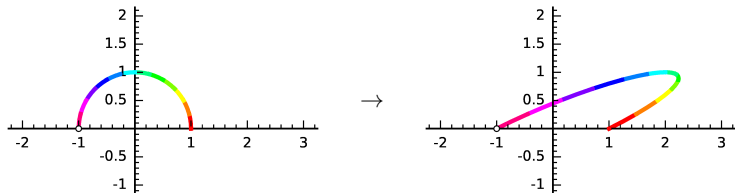


The two dilations combine independently in that the first coordinate of the output uses only x and the second coordinate of the output uses only y .

Skew

Example Next is a map with an output coordinate affected by both x and y .

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + 2y \\ y \end{pmatrix}$$

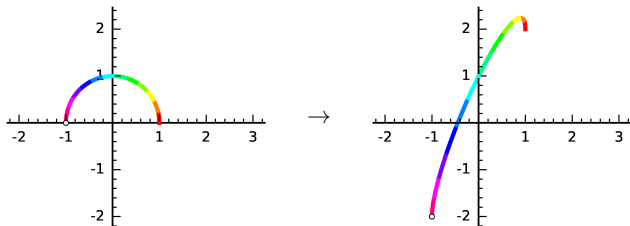


On the x axis, where $y = 0$, the output's first coordinate is the same as the input's first coordinate. However as we move away from the x axis the y 's get larger, and the first coordinate of the output is increasingly affected. (One definition of *skew* is: having an oblique direction or position; slanting.)

Skew the other way

Example We can flip which output coordinate is affected by both x and y .

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ 2x + y \end{pmatrix}$$

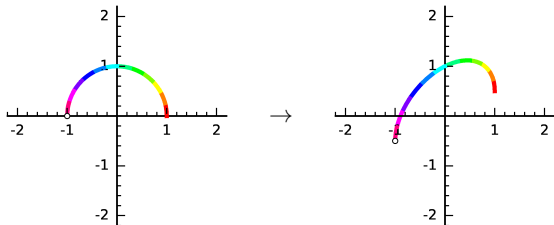


In addition to dilation we see clear rotation, for instance of the red input vector.

Same idea but with a smaller effect

Example

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ (1/2)x + y \end{pmatrix}$$



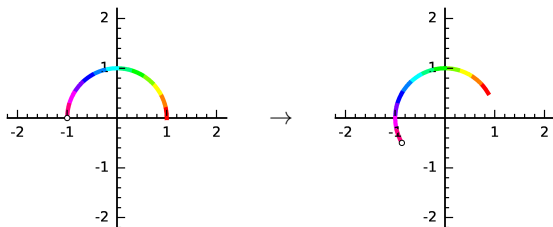
Observe that the rotation is not even. A red vector is rotated quite a bit but a green vector, near the y axis, is not rotated much. And right on the y-axis the vector is not rotated at all.

Pure roation

Example This rotates every vector counterclockwise through the angle θ .

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} \cos(\theta) \cdot x - \sin(\theta) \cdot y \\ \cos(\theta) \cdot x + \sin(\theta) \cdot y \end{pmatrix}$$

In this picture $\theta = \pi/6$.

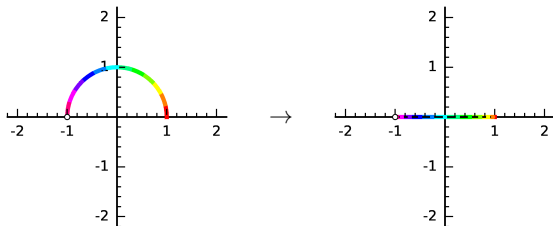


Projection

Example Maps can lose a dimension.

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ 0 \end{pmatrix}$$

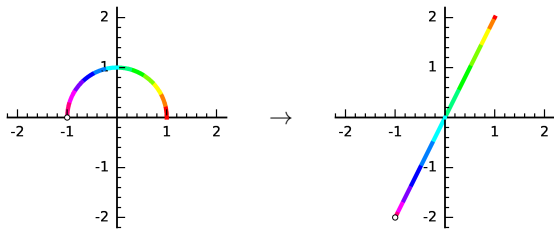
The output is one-dimensional.



Example The map may project the input vector to a line that is not an axis.

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ 2x \end{pmatrix}$$

The two-dimensional input is sent to an output that is one-dimensional.

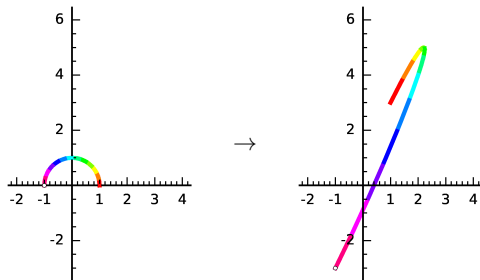


A generic map

Example An arbitrary map

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}$$

may have an action that is a mixture of the effects shown above.



This shows dilation, rotation, and orientation reversal.